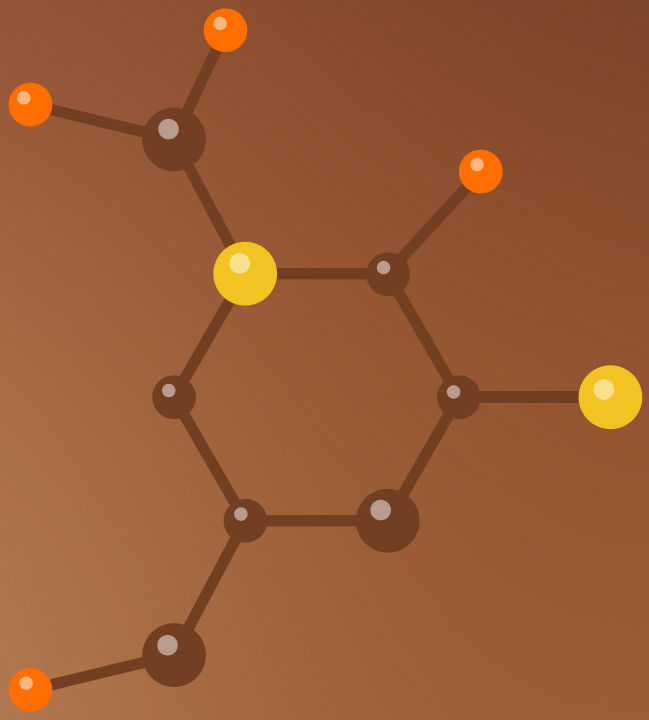
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KOTA (RAJASTHAN)



# ORGANIC CHEMISTRY

## JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

**EXERCISE**

## Chemistry in Everyday Life

English Medium

## EXERCISE # O-I

**1.** An antipyretic is –
(A) Seldane (B) Paracetamol (C) Luminal (D) Aspartame
**CD0001**

**2.** Medicine which is an antibiotic is –
(A) Ampicillin (B) Aspirin (C) Dimetapp (D) None of these
**CD0002**

**3.** Paracetamol is –
(A) Both antipyretic and analgesic (B) Analgesic
(C) Antipyretic (D) Antimalarial
**CD0003**

**4.** Sulpha drugs are derivatives of –
(A) Benzene sulphonic acid (B) Sulphanillic acid
(C) Sulphanilamide (D) p - aminobenzoic acid
**CD0004**

**5.** Aspirin is called –
(A) Pyretic (B) Antiseptic (C) Antibiotic (D) Antipyretic
**CD0005**

**6.** Which of the following is an antidiabetic drug –
(A) Insulin (B) Penicillin (C) Sucralose (D) Aspirin
**CD0006**

**7.** 2-Acetoxybenzoic acid is called –
(A) Antiseptic (B) Aspirin (C) Antibiotic (D) Mordant dye
**CD0007**

**8.** Arsenic drugs are mainly used in the treatment of–
(A) Jaundice (B) Typhoid (C) Syphilis (D) Cholera
**CD0008**

**9.** Aspirin is an acetylation product of –
(A) p-dihyroxybenzene (B) o-hydroxybenzoic acid
(C) o-dihydroxy benzene (D) m-hydroxybenzoic acid
**CD0009**

**10.** Chloramphenicol is an –
(A) Analgesic (B) Anaesthetic (C) Antibiotic (D) Antiseptic
**CD0010**

**11.** Substances which bring body temperature down are known as –
(A) Antipyretics (B) Analagin (C) Antibiotics (D) Hypnotics
**CD0011**

**12.** The drug given during hypertension is –
(A) Salvarsan (B) Chloroxylenol (C) Equanil (D) Aspirin
**CD0012**

**13.** Which of the following is known as narrow spectrum antibiotic –
(A) Ofloxacin (B) Ampicillin (C) Chloramphenicol (D) Penicillin
**CD0013**

**14.** Phenol is used as –
(A) An antiseptic (B) A disinfectant (C) Both (A) and (B) (D) None of these
**CD0014**

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**15.** Mixture or chloroxylenol and terpineol acts as
(A) antiseptic (B) antipyretic (C) antibiotic (D) analgesic

**CD0015**

**16.** Bithional is generally added to the soaps as an additive to function as a/an
(A) buffering agent (B) antiseptic (C) softener (D) dryer.

**CD0016**

**17.** Artificial sweetner which is stable under only cold conditions is
(A) saccharine (B) sucralose (C) aspartame (D) alitame.

**CD0017**

**18.** Antiseptics and disinfectants either kill or prevent growth of mierorganisms. Identify which of the following statements is not true.
(A) Dilute solution of boric acid is strong antiseptic.
(B) Disinfectants harm the living tissues.
(C) A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant.
(D) Chlorine and iodine are used as strong disinfectants.

**CD0018**

**19.** Dettol is the mixture of
(A) chloroxylenol and bithional (B) chloroxylenol and terpineol
(C) phenol and iodine (D) terpineol and bithional

**CD0019**

**20.** Chloramphenicol is an
(A) antifertility drug (B) antihistamine
(C) antiseptic and disinfectant (D) antibiotic-broad spectrum

**CD0020**

**21.** Which of the following forms cationic micelles above certain concentration?
(A) Sodium dodecyl sulphate (B) Sodium acetate
(C) Urea (D) Cetyltrimethylammonium bromide.

**CD0021**

**22.** Aspirin is an acetylation product of
(A) m-Hydroxybenzoic acid (B) o-Dihydroxybenzene
(C) o-Hydroxybenzoic acid (D) p-Dihydroxybenzene

**CD0022**

**23.** Which of the following can possibly be used as analgesic without causing addiction and mood modification?
(A) Codeine (B) Heroin
(C) Morphine (D) N-Acetyl-para-aminophenol.

**CD0023**

**24.** Diazo coupling is useful to prepare some
(A) pesticides (B) dyes (C) proteins (D) vitamins.

**CD0024**

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**25.** Which one of the following statements is not true?
(A) Penicillin is a natural antibiotic
(B) Aspirin is both analgesic and antipyretic
(C) Sulphadiazine is a synthetic antibacterial drug
(D) Some disinfectants can be used as antiseptics.

**CD0025**

**26.** Which of the following is a correct match?
(A) Anionic detergent : Glyceryl ester of stearic acid
(B) Fat : Sodium stearate
(C) Soap : Sodium salt of oleic acid
(D) Soap : Sodium Lauryl Sulphate

**CD0026**

**27.** Which of the following compounds is not a food preservative?
(A) Table salt (B) Sugar
(C) Vegetable oil (D) Sodium oleate

**CD0027**

**28.** What is the normal range of molecular masses of drugs?
(A) (~0–100) u (B) (~100–500) u
(C) (~500-1000) u (D) > 1000 u

**CD0028**

**29.** Which of the following is not used as an anti-oxidant?
(A) Butylated Hydroxy Toluene (BHT)
(B) Sulphur Dioxide
(C) Sulphite
(D) Sodium Palmate

**CD0029**

**30.** Abrasive is a :
(A) Scouring agent (B) Drug to reduce fever
(C) Drug to kill bacteria (D) Gum

**CD0030**

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## EXERCISE # O-II

**1.** Which of the following are used as analgesics?
(A) Aspirin (B) Heroin
(C) Promethazine (D) Serotonin

**CD0031**

**2.** Select the correct statements :
(A) Drugs are chemicals of low molecular mass.
(B) Drugs produce biological response.
(C) Drugs which are used in diagnosis, prevention and treatment of diseases are called medicines.
(D) Chemotherapy is the use of chemical for therapeutic effect.

**CD0032**

**3.** Which of the following pairs are bacteriostatic antibiotics?
(A) Penicillin, tetracycline
(B) Erythromycin chloramphenicol
(C) Ofloxacin, aminoglycosides
(D) Tetracycline, chloramphenicol

**CD0033**

**4.** Which of the following can be used as artificial sweetners?
(A) Aspartame (B) Alitame
(C) Sucralose (D) Saccharin

**CD0034**

**5.** Which of the following is known as broad spectrum antibiotic –
(A) Streptomycin (B) Ampicillin
(C) Chloramphenicol (D) Penicillin

**CD0035**

**6.** Medicine which is an antibiotic is –
(A) Ampicillin (B) Ofloxacin (C) Aminoglycoside (D) Penicillin

**CD0036**

**7.** Bactericidal antibiotics are -
(A) Penicilline (B) Ofloxacin
(C) Aminoglycosides (D) Only (B) and (C)

**CD0037**

**8.** Bacteristatic antibiotics are -
(A) Chloramphenicol (B) Tetracydine
(C) Penicillin (D) Erythromycin

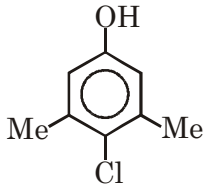
**CD0038**

**9.** Which one drugs have one or more then one chiral atom -
(A) Penicillin
(B) Chloramphenicol
(C) Terpineol
(D) Phenelzine (Nardil)

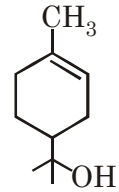
**CD0039**

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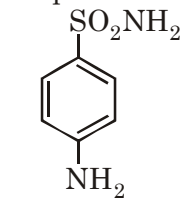
**10.** How many drugs have even number of degree of unsaturation -

(A) 

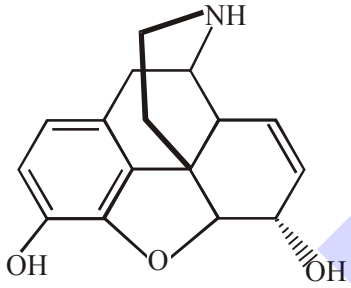
Chloroxylenol

(B) 

Terpineol

(C) 

Sulphanilaimide

(D) 

Morphine

**CD0040**

**11.** Which pair is combination of non narcotic and narcotic drug?

(A) Paracetamol, Aspirin (B) Paracetamol, Herion

(C) Aspirin, Morphine (D) Morphine, Codeine

**CD0041**

**12.** Which one is/are used as food preservative?

(A) Sodium benzoate (B) Vegetable oil (C) Sorbic acid (D) Propanoic acid

**CD0042**

**13.** Choose the incorrect statement?

(A) Saccharin is 650 times sweeter than cane sugar.

(B) Alitame is 2000 times sweeter than cane sugar

(C) Sucralose is 160 times sweeter than cane sugar.

(D) Aspartame is 550 times sweeter than cane sugar.

**CD0043**

**14.** Antiseptic chloroxylenol is :

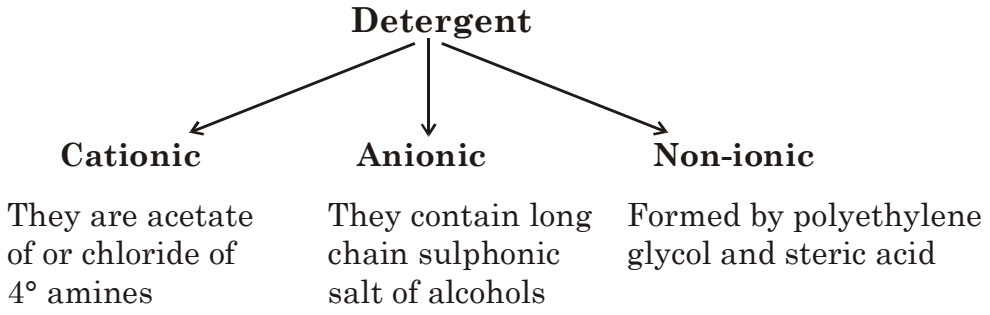
(A) 4-chloro-3,5-dimethyl phenol (B) 3-chloro-4, 5-dimethyl phenol

(C) 4-chloro-2, 5-dimethyl phenol (D) 5-chloro-3,4-dimethyl phenol

**CD0044**

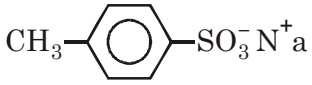
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**Paragraph for Question No. 15 to 17**

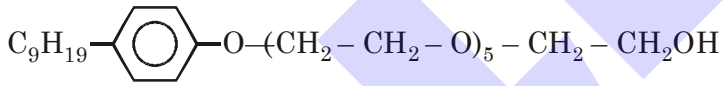



**15.** Used in liquid dishwashing detergent?

(A) $CH_3(-CH_2)_{15}-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{N^+}}-CH_3Br^-$

(B) 

(C) $CH_3(CH_2)_{10}CH_2OSO_3^-\overset{+}{N}a$

(D) 

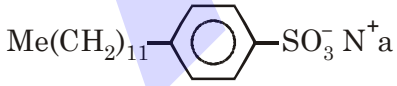
**CD0045**

**16.** Which is not used in synthetic detergent?

(A) $(C_{17}H_{35}COO)_2Ca$

(B) $CH_3(CH_2)_{16}CH_2OSO_3^-\overset{+}{N}a$

(C) $Me-(CH_2)_{15}\overset{+}{N}MeCl^-$

(D) 

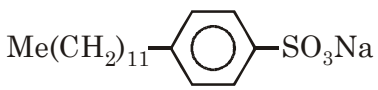
**CD0046**

**17.** Which of the following is non-ionic detergent?

(A) $Me(CH_2)_{16}COO(CH_2CH_2O)_3 - CH_2CH_2OH$

(B) $Me(CH_2)_{16}CH_2SO_3Na$

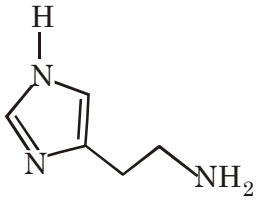
(C) $Me(CH_2)_{15} - NMe_3Br$

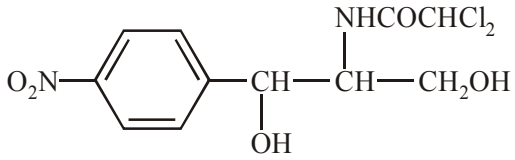
(D) 

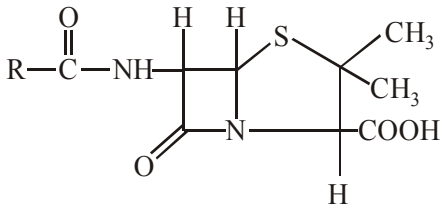
**CD0047**

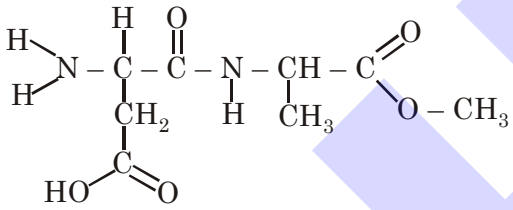
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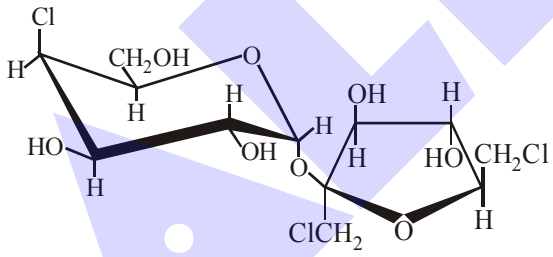
**18.** How many organic compound(s) is/are correctly matched with number of chiral carbon(s) present in it's structure :

(i)  Histamine = 0

(ii)  Chloramphenicol = 2

(iii)  Penicillin = 3

(iv)  Aspartame = 4

(v)  Sucralose = 9

**CD0048**

**Matrix match :**

**19.**

| **Column-I** | **Column-II** |
|---|---|
| (A) Asprin | (P) Antiseptic and Disinfectant |
| (B) Paracetamol | (Q) Analgesic |
| (C) Soframycin | (R) Antibiotic |
| (D) Pencillin | (S) Antipyretic |

**CD0049**

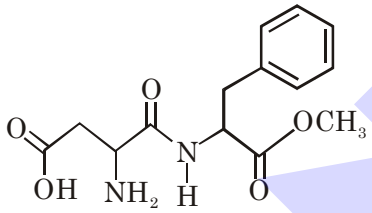
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**20.** Match list-I and list-II and select the correct answer using the codes given below the list?

| **Column-I** | **Column-II** |
|---|---|
| (A) Penicillin | (P) Antiseptic |
| (B) Chloramphenicol | (Q) Antibiotic |
| (C) 0.2 % solution of phenol | (R) Bacteriocidal |
| (D) 1 % solution of phenol | (S) Disinfectant |

**CD0050**

**21.** "Aspartame" is roughly 100 times as sweet as cane sugar. On complete hydrolysis of aspartame, products obtained is/are :



(A) $PhCH_2-\underset{\displaystyle CO_2H}{\underset{|}{CH}}-NH_2$ (B) $H_2N-\underset{\displaystyle CH_2CO_2H}{\underset{|}{CH}}-CO_2H$ (C) $CH_3OH$ (D) $CH_3-\underset{\displaystyle CO_2H}{\underset{|}{CH}}-NH_2$

**BM0045**

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## EXERCISE # JEE MAIN'S

**1.** Which one of the following types of drugs reduces fever- **[AIEEE-2005]**

(1) Tranquilizer (2) Antibiotic (3) Antipyretic (4) Analgesic

**CD0051**

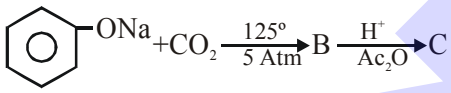
**2.** Aspirin is known as :- **[AIEEE-2012]**

(1) Methyl salicylic acid (2) Acetyl salicylic acid

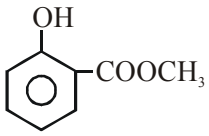
(3) Phenyl salicylate (4) Acetyl salicylate

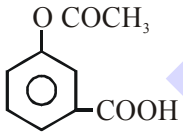
**CD0052**

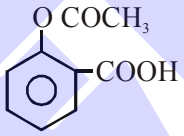
**3.** Sodium phenoxide when heated with $CO_2$ under pressure at 125°C yields a product which on acetylation produces C. **[J-Main- 2014]**

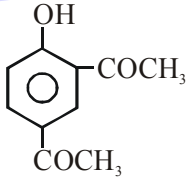



The major product C would be :

(1) 


(2) 


(3) 


(4) 

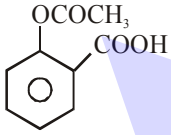

**CD0053**

**4.** Which of the following compounds is not an antacid? **[J-Main- 2015]**

(1) Phenelzine (2) Ranitidine

(3) Aluminium hydroxide (4) Cimetidine

**CD0054**

**5.** 

is used as : **[J-Main-2015]**

(1) Insecticide (2) Antacid (3) Antihistamine (4) Analgesic

**CD0055**

**6.** Which of the following is an anionic detergent ? **[J-Main-2016]**

(1) Glyceryl oleate

(2) Sodium stearate

(3) Sodium lauryl sulphate

(4) Cetyltrimethyl ammonium bromide

**CD0056**

**7.** The artificial sweetener that has the highest sweetness value in comparison to cane sugar is:

(1) Saccharin (2) Alitame **[J-Main-2016]**

(3) Aspartane (4) Sucralose

**CD0057**

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**8.** The correct match between items of List-I and List-II is :- **[J-Main-2018]**

| | **Litst-I** | | **List-II** |
|---|---|---|---|
| (A) | Phenelzine | (P) | Pyrimidine |
| (B) | Chloroxylenol | (Q) | Furan |
| (C) | Uracil | (R) | Hydrazine |
| (D) | Ranitidine | (S) | Phenol |

(1) (A)-(S), (B)-(R), (C)-(P), (D)-(Q) (2) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
(3) (A)-(S), (B)-(R), (C)-(Q), (D)-(P) (4) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)

**BM0129**

**9.** The correct match between Item(I) and Item(II) is : **[JEE-MAIN-(Jan)-2019]**

| **Item-I** | **Item-II** |
|---|---|
| (A) Norethindrone | (P) Anti-biotic |
| (B) Ofloxacin | (Q) Anti-fertility |
| (C) Equanil | (R) Hypertension |
| | (S) Analgesics |

(1) A-R, B-P, C-S (2) A-Q, B-P, C-R (3) A-R, B-P, C-R (4) A-Q, B-R, C-S

**CD0058**

**10.** The correct match between Item -I and Item-II is : **[JEE-MAIN-(Jan)-2019]**

| **Item – I (drug)** | | **Item – II (test)** | |
|---|---|---|---|
| **(A)** | Chloroxylenol | **(P)** | Carbylamine Test |
| **(B)** | Norethindrone | **(Q)** | Sodium Hydrogen carbonateTest |
| **(C)** | Sulphapyridine | **(R)** | Ferric chloride test |
| **(D)** | Penicillin | **(S)** | Bayer's test |

(1) A→Q ; B→P ; C→S ; D→R (2) A→R ; B→P ; C→S ; D→Q
(3) A→R ; B→S ; C→P ; D→Q (4) A→Q ; B→S ; C→P ; D→R

**CD0059**

**11.** Noradrenaline is a /an **[JEE-MAIN-(April)-2019]**

(1) Neurotransmitter (2) Antidepressant (3) Antihistamine (4) Antacid

**CD0060**

**12.** The number of $sp^2$ hybridised carbons present in "Aspartame" is _____. **[JEE-MAIN-(Jan)-2020]**

**CD0061**

**13.** The number of chiral carbons in chloramphenicol is _____ . **[JEE-MAIN-(Jan)-2020]**

**CD0062**

**14.** The number of chiral centres in penicillin is _____. **[JEE-MAIN-(Jan)-2020]**

**CD0063**

**15.** The mass percentage of nitrogen in histamine is _____. **[JEE-MAIN-(Jan)-2020]**

**CD0064**

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**16.** The gas released during anaerobic degradation of vegetation may lead to : **[JEE-MAIN-2021]**

(1) Ozone hole
(2) Acid rain
(3) Corrosion of metals
(4) Global warming and cancer

**CD0065**

**17.** Match List-I and List-II. **[JEE-MAIN-2021]**

| | **List-I** | | **List-II** |
|---|---|---|---|
| (a) | Valium | (i) | Antifertility drug |
| (b) | Morphine | (ii) | Pernicious anaemia |
| (c) | Norethindrone | (iii) | Analgesic |
| (d) | Vitamin $B_{12}$ | (iv) | Tranquilizer |

(1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)
(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)
(3) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)
(4) (a)–(i), (b)–(iii), (c)–(iv), (d)–(ii)

**CD0066**

**18.** The presence of ozone in troposphere **[JEE-MAIN-2021]**

(1) Protects us from the UV radiation
(2) Protects us from the X-ray radiation
(3) Protects us from greenhouse effect
(4) generates photochemical smog

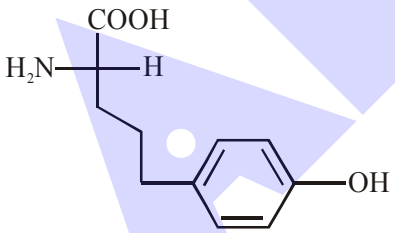
**CD0067**

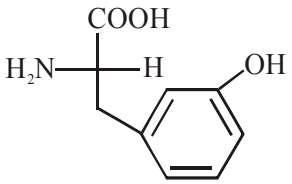
**19.** The functions of antihistamine are : **[JEE-MAIN-2021]**

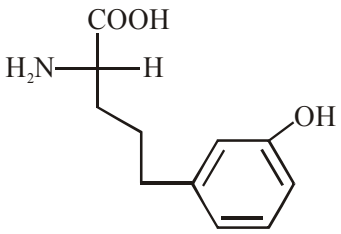
(1) Antiallergic and Analgesic
(2) Antacid and antiallergic
(3) Analgesic and antacid
(4) Antiallergic and antidepressant

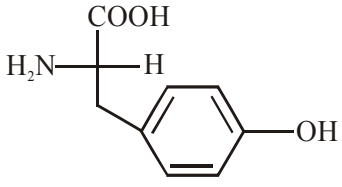
**CD0068**

**20.** Which of the following is correct structure of tyrosine? **[JEE-MAIN-2021]**

(1) 


(2) 


(3) 


(4) 


**CD0069**

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**21.** Match List-I with List-II : **[JEE-MAIN-2021]**

| | **List-I Chemical Compound** | | **List-II Used as** |
|---|---|---|---|
| (a) | Sucralose | (i) | Synthetic detergent |
| (b) | Glyceryl ester of stearic acid | (ii) | Artificial sweetener |
| (c) | Sodium benzoate | (iii) | Antiseptic |
| (d) | Bithionol | (iv) | Food preservative |

Choose the correct match :

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) (4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

**CD0070**

**22.** Which of the following statement(s) is (are) incorrect reason for eutrophication ?

(A) excess usage of fertilisers **[JEE-MAIN-2021]**

(B) excess usage of detergents

(C) dense plant population in water bodies

(D) lack of nutrients in water bodies that prevent plant growth

Choose the most appropriate answer from the options given below :

(1) (A) only (2) (C) only (3) (B) and (D) only (4) (D) only

**CD0071**

**23.** Match the list -I with list - II **[JEE-MAIN-2021]**

| | **List-I (Class of Drug)** | | **List-II (Example)** |
|---|---|---|---|
| (a) | Antacid | (i) | Novestrol |
| (b) | Artificial sweetener | (ii) | Cimetidine |
| (c) | Antifertility | (iii) | Valium |
| (d) | Tranquilizers | (iv) | Alitame |

(1) (a) – (ii), (b) – (iv),(c) – (i), (d) – (iii) (2) (a) – (iv), (b) – (i),(c) – (ii), (d) – (iii)

(3) (a) – (iv), (b) – (iii),(c) – (i), (d) – (ii) (4) (a) – (ii), (b) – (iv),(c) – (iii), (d) – (i)

**CD0072**

**24.** Match List-I with List-II **[JEE-MAIN-2021]**

| List-I | List-II |
|---|---|
| (a) Chlorophyll | (i) Ruthenium |
| (b) Vitamin-$B_{12}$ | (ii) Platinum |
| (c) Anticancer drug | (iii) Cobalt |
| (d) Grubbs catalyst | (iv) Magnesium |

Choose the most appropriate answer from the options given below :

(a) a-iii, b-ii, c-iv, d-i (b) a-iv, b-iii), c-ii, d-i

(c) a-iv, b-iii, c-i, d-ii (d) a-iv, b-ii, c-iii, d-i **CD0073**

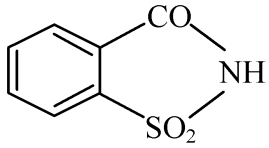
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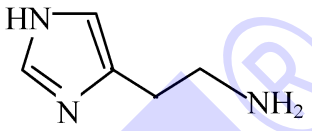
**25.** Match List - I with List - II : **[JEE-MAIN-2021]**

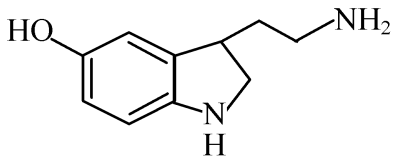
| **List - I (Class of Chemicals)** | **List - II (Example)** |
|---|---|
| (a) Antifertility drug | (i) Meprobamate |
| (b) Antibiotic | (ii) Alitame |
| (c) Tranquilizer | (iii) Norethindrone |
| (d) Artificial Sweetener | (iv) Salvarsan |

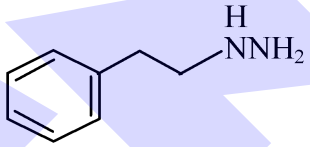
(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii) **CD0074**

**26.** Which one of the following chemicals is responsible for the production of HCl in the stomach leading to irritation and pain? **[JEE-MAIN-2021]**

(1) 

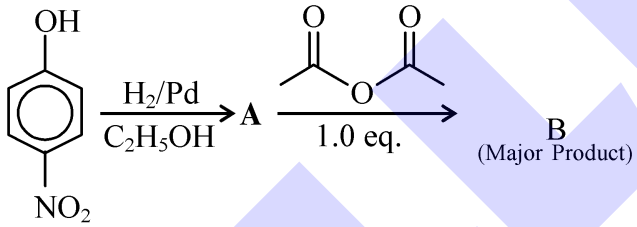
(2) 

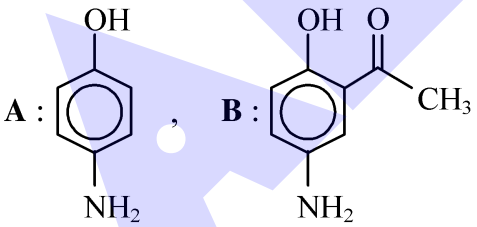
(3) 

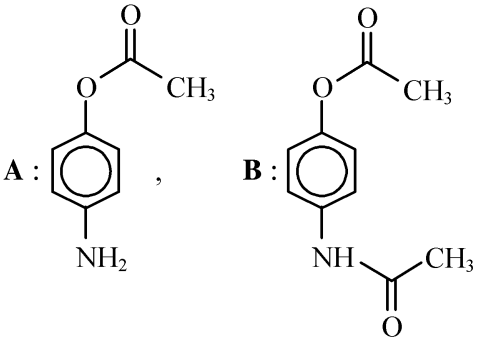
(4) 

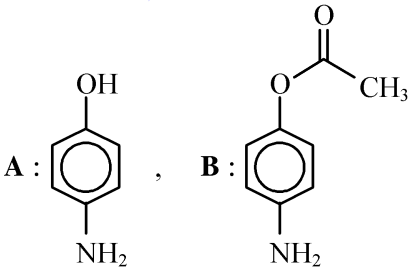
**CD0075**

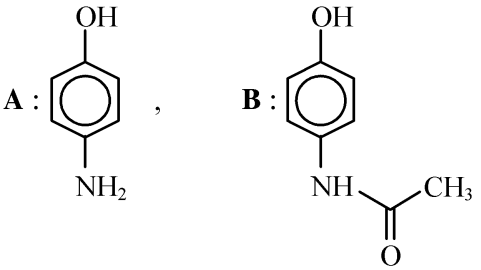
**27.** The correct structures of A and B formed in the following reactions are : **[JEE-MAIN-2021]**



(1) 

(2) 

(3) 

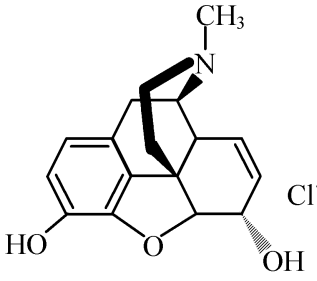
(4) 

**CD0076**

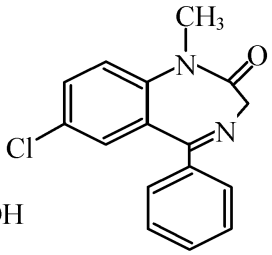


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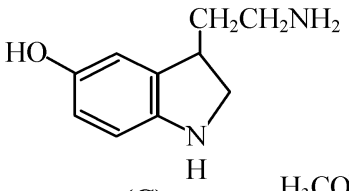
**28.** **[JEE-MAIN-2021]**



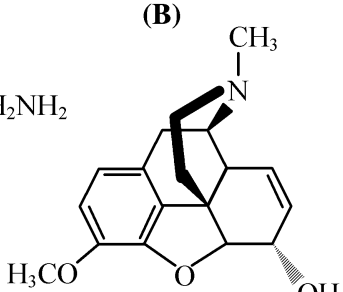

**(A)**




**(B)**




**(C)**




**(D)**

The correct statement about (A), (B), (C) and (D) is :

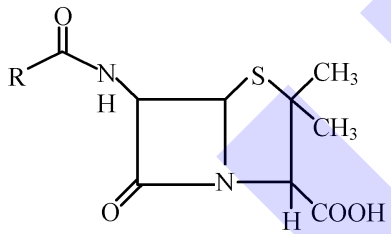
(1) (A), (B) and (C) are narcotic analgesics
(2) (B), (C) and (D) are tranquillizers
(3) (A) and (D) are tranquillizers
(4) (B) and (C) are tranquillizers

**CD0077**

**29.** Given below are two statements : **[JEE MAIN 2021]**

Statement I : Penicillin is a bacteriostatic type antibiotic.

Statement II : The general structure of Penicillin is:




Choose the correct option :

(1) Both statement I and statement II are false
(2) Statement I is incorrect but statement II is true
(3) Both statement I and statement II are true
(4) Statement I is correct but statement II is false

**CD0078**

**30.** Match List-I with List-II : **[JEE MAIN 2021]**

| **List-I (Drug)** | **List-II (Class of Drug)** |
|---|---|
| (a) Furacin | (i) Antibiotic |
| (b) Arsphenamine | (ii) Tranquilizers |
| (c) Dimetone | (iii) Antiseptic |
| (d) Valium | (iv) Synthetic antihistamines |

Choose the most appropriate match :

(1) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

**CD0079**

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**31.** With respect to drug-enzyme interaction, identify the wrong statement: **[JEE MAIN 2021]**

(1) Non-Competitive inhibitor binds to the allosteric site

(2) Allosteric inhibitor changes the enzyme's active site

(3) Allosteric inhibitor competes with the enzyme's active site

(4) Competitive inhibitor binds to the enzyme's active site

**BM0198**

**32.** Which of the following is not a broad spectrum antibiotic? **[JEE MAIN 2022]**

(1) Vancomycin (2) Ampicillin

(3) Ofloxacin (4) Penicillin G **CD0080**

**33.** Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes because **[JEE MAIN 2022]**

(1) soap particles remain floating in water as ions

(2) the hydrophobic part of soap is not able to take away grease

(3) the micelles are not formed due to concentration of soap, below its CMC value

(4) colloidal structure of soap in water is completely disturbed. **CD0081**

**34.** Which one of the following is an example of artificial sweetner ? **[JEE MAIN 2022]**

(1) Bithional (2) Alitame (3) Salvarsan (4) Lactose

**CD0082**

**35.** Match List I with List II. **[JEE MAIN 2022]**

| List I | List II |
|---|---|
| A. Zymase | I. Stomach |
| B. Diastase | II. Yeast |
| C. Urease | III. Malt |
| D. Pepsin | IV. Soyabean |

Choose the correct answer from the options given below:

(1) A-II, B-III, C-I, D-IV (2) A-II, B-III, C-IV, D-I

(3) A-III, B-II, C-IV, D-I (4) A-III, B-II, C-I, D-IV **CD0083**

**36.** A chemical which stimulates the secretion of pepsin is : **[JEE MAIN 2022]**

(1) Anti histamine (2) Cimetidine (3) Histamine (4) Zantac

**CD0084**

**37.** Match List I with List II **[JEE MAIN 2022]**

| List I | List II |
|---|---|
| A. Antipyretic | I. Reduces pain |
| B. Analgesic | II. Reduces stress |
| C. Tranquilizer | III. Reduces fever |
| D. Antacid | IV. Reduces acidity (Stomach) |

Choose the correct answer from the options given below:

(1) A-III, B-I, C-II, D-IV (2) A-III, B-I, C-IV, D-II

(3) A-I, B-IV, C-II, D-III (4) A-I, B-III, C-II, D-IV **CD0085**

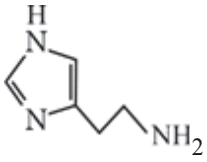
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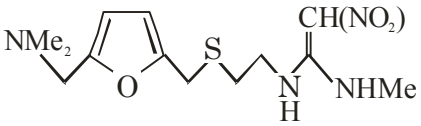
**38.** Which amongst the following is not a pesticide ? **[JEE MAIN 2022]**

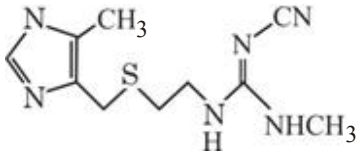
(1) DDT (2) Organophosphates (3) Dieldrin (4) Sodium arsenite

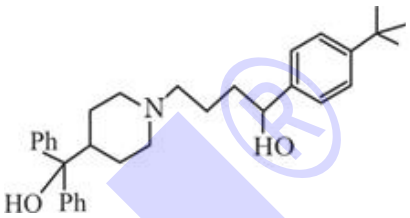
**CD0086**

**39.** The drug tegamet is : **[JEE MAIN 2022]**

(1) 

(2) 

(3) 

(4) 

**CD0087**

**40.** The mixture of chloroxylenol and terpineol is an example of : **[JEE MAIN 2022]**

(1) antiseptic (2) pesticide (3) disinfectant (4) narcotic analgesic

**CD0088**

**41.** The acid that is believed to be mainly responsible for the damage of Taj Mahal is **[JEE MAIN 2022]**

(1) Sulfuric acid (2) Hydrofluoric acid (3) Phosphoric acid (4) Hydrochloric acid

**CD0089**

**42.** Which of the following compound does not contain sulphur atom ? **[JEE MAIN 2022]**

(1) Cimetidine (2) Ranitidine (3) Histamine (4) Saccharin

**CD0090**

**43.** Match List I with List II. **[JEE MAIN 2022]**

| List–I | | List–II | |
|---|---|---|---|
| A. | Sulphate | I. | Pesticide |
| B. | Fluoride | II. | Bending of bones |
| C. | Nicotine | III. | Laxative effect |
| D. | Sodium arsinite | IV. | Herbicide |

Choose the correct answer from the options given below:

(1) A-II, B-III. C-IV, D-I (2) A-IV, B-III, C-II, D-I

(3) A-III, B-II, C-I, D-IV (4) A-III, B-II, C-IV, D-I **CD0091**

**44.** Some drugs bind to a site other than, the active site of an enzyme. This site is known as **[JEE MAIN 2022]**

(1) non-active site (2) allosteric site (3) competitive site (4) therapeutic site

**CD0092**

**45.** Drugs used to bind to receptors, inhibiting its natural function and blocking a message are called :

(1) Agonists (2) Antagonists **[JEE MAIN 2022]**

(3) Allosterists (4) Anti histaminists **CD0093**

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**46.** Which among the following pairs has only herbicides ? **[JEE MAIN 2022]**

(1) Aldrin and Dieldrin (2) Sodium chlorate and Aldrin

(3) Sodium arsinate and Dieldrin (4) Sodium chlorate and sodium arsinite.

**CD0094**

**47.** Which of the following compounds is an example of hypnotic drug ? **[JEE MAIN 2022]**

(1) Seldane (2) Amytal (3) Aspartame (4) Prontosil

**CD0095**

**48.** Given below are two statements: **[JEE MAIN 2022]**

Statement I : The non bio-degradable fly ash and slag from steel industry can be used by cement industry.

Statement II : The fuel obtained from plastic waste is lead free.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct **CD0096**

**49.** Statements about Enzyme Inhibitor Drugs are given below : **[JEE MAIN 2022]**

(A) There are Competitive and Non-competitive inhibitor drugs.

(B) These can bind at the active sites and allosteric sites.

(C) Competitive Drugs are allosteric site blocking drugs.

(D) Non-competitive Drugs are active site blocking drugs.

Choose the correct answer from the options given below :

(1) (A), (D) only (2) (A), (C) only (3) (A), (B) only (4) (A), (B), (C) only

**CD0097**

**50.** Two statements in respect of drug-enzyme interaction are given below **[JEE MAIN 2022]**

Statement I : Action of an enzyme can be blocked only when an inhibitor blocks the active site of the enzyme.

Statement II : An inhibitor can form a strong covalent bond with the enzyme.

In the light of the above statements. Choose the correct answer from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true **CD0098**

**Integer**

**51.** In alanylglycylleucylalanylvaline, the number of peptide linkages is_____. **[JEE MAIN 2022]**

**CD0099**

**52.** The number of chlorine atoms in bithionol is____. **[JEE MAIN 2022]**

**CD0100**

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# ANSWER KEY

## EXERCISE # O-I

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | A | A | C | D | A | B | C | B | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | C | D | C | A | B | C | A | B | D |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | B | C | D | B | A | C | D | B | D | A |

## EXERCISE # O-II

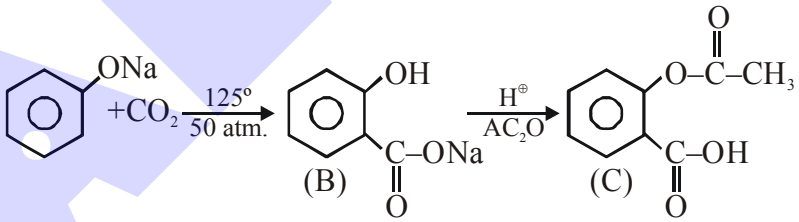
| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A,B | A,B,C,D | B,D | A,B,C,D | B,C | A,B,C,D | A,B,C | A,B,D | A,B | A,B,C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | |
| Ans. | B,C | A,B,C,D | A,C,D | A | D | A | A | 4 | A→Q ; B→S ; C→P ; D→R | |
| Que. | 20 | | 21 | | | | | | | |
| Ans. | A→Q,R ; B→Q ; C→P ; D→S | | A,B,C | | | | | | | |

## EXERCISE # JEE MAIN'S

**1. (3)** **2. (2)**

**3. (3)**

**Sol.** First step is carboxylation (Kolbe schmidt reaction) & second step is acetylation of sodium salt of aspirin (B).

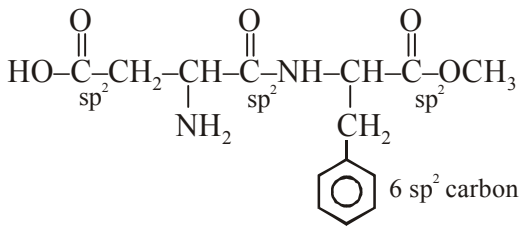



**4. (1)** **5. (4)** **6. (3)** **7. (2)** **8. (2)** **9. (2)**

**10. (3)** **11. (1)**

**12. (9.00)**

**Sol.**

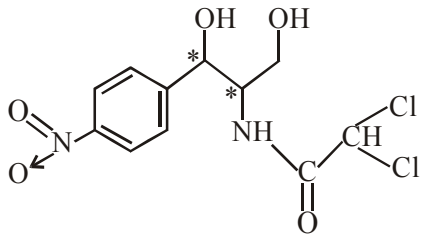



no. of $sp^2$–carbon → 9

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**13. (2.00)**

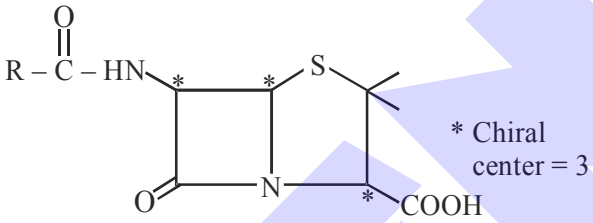
**Sol.**



Chloramphenicol

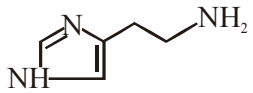
**14. (3.00)**

**Sol.** The structure of penicillin is




**15. (37.80 to 38.20)**

**Sol.**



M.F. of Histamine is $C_5H_9N_3$

Molecular mass of Histamine is 111

Now, mass % of nitrogen = $\left(\frac{42}{111}\right)\times 100 = 37.84\%$

| | | | | | |
|---|---|---|---|---|---|
| **16. (4)** | **17. (2)** | **18. (4)** | **19. (2)** | **20. (4)** | **21. (2)** |
| **22. (4)** | **23. (1)** | **24. (2)** | **25. (3)** | **26. (2)** | **27. (4)** |
| **28. (4)** | **29. (2)** | **30. (4)** | **31. (3)** | | |

**32. (4)**

**Sol.** Penicillin G following is a narrow spectrum antibiotic

**33. (3)**

**Sol.** Micelle formation only takes place above CMC.

**34. (2)**

**Sol.** Alitame is a second generation dipeptide sweetner that is 200 times sweeter than sucrose.

**35. (2)**

**Sol.** Zymase naturally occurs in yeast.

Diastase is found in malt.

Urease is found in soyabean

Pepsin is found in stomach



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**36. (3)**

**Sol.** Histamine (It is use for secretion of pepsin & HCl in stomach)

**37. (1)**

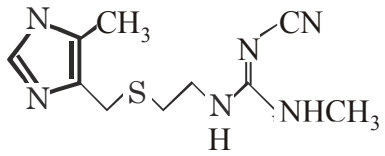
**Sol.**

| | |
|---|---|
| A. Antipyretic | Reduces fever |
| B. Analgesic | Reduces pain |
| C. Tranquilizer | Reduces stress |
| D. Antacid | Reduces acidity (Stomach) |

**38. (4)**

**39. (3)**

**Sol.** Tegamet is the brand name of Cimetidine



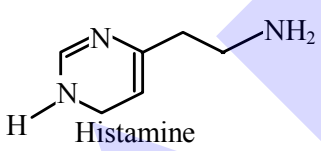
**40. (1)**

**Sol.** Antiseptic Dettol is mixture of chloroxylenol and terpineol.

**41. (1)**

**Sol.** $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$

**42. (3)**




Histamine is nitrogenous compound it does not contain sulpher.

**43. (3)**

**Sol.** A-Sulphate - III (Laxative effect)

B-Fluoride - II (Bending of bones)

C-Nictoine - I (pesticides)

D-Sodium Arsinite - IV (herbicide)

**44. (2)**

**Sol.** Theoretical

**45. (2)**

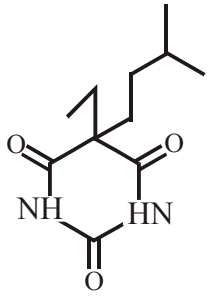
**46. (4)**

**Sol.** Both sodium chlorate and sodium arsenite behave as herbicide.

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**47. (2)**

Sol. Amytal is hypnotic drug used to treat sleeping disorder.




Amytal

**48. (1)**

**49. (3)**

**Sol.** Enzyme inhibitors can be competitive inhibitors (inhibit the attachment of substrate on active site of enzyme) and non-competitive inhibitor (changes the active site of enzyme after binding at allosteric site.)

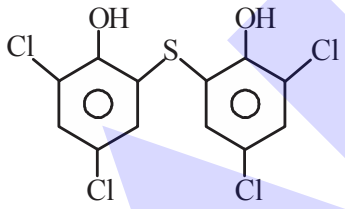
**50. (4)**

**51. (4)**

**Sol.** There are Five amino acids and four peptide linkages.

**52. (4)**

Sol. Bithinol




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